Symmetry Groups of the Plane

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1 Introduction

Group theory and geometry are closely intertwined fields of mathematics which provide powerful tools for systematizing the study of certain concepts which map very well onto practical, real-world problems. One such concept with rich potential for study in both fields is **symmetry**, a property which broadly applies to objects which are invariant or "the same" under certain kinds of transformations.

This paper will begin with a brief overview of some basic concepts from group theory, followed by an exploration of the application of group theory to the study of symmetrical objects in the Euclidean plane. Symmetry groups are an excellent and succinct way to classify certain *ways* in which an object can exhibit symmetry without needing to be concerned with the particulars of the object's shape. Symmetry groups allow us to describe the fact that a square, the letter "X" (when written to occupy a square), and the flag of New Mexico share the same patterns of symmetry (more specifically D_4 symmetry, as we will identify later) without needing to concern ourselves with the specific manner in which each shape is constructed.

2 Introduction to groups

Definition 1. *A **group** is a set G equipped with a binary operation $\circ : G \to G$ such that*

- *there exists an identity element e such that $e \circ a = a \circ e = a$ for all $a \in G$;*
- *$\circ$ is associative: $a \circ (b \circ c) = (a \circ b) \circ c$ for all $a, b, c \in G$; and*
- *for all $a \in G$ there exists $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.*

Often $\circ$ is omitted for brevity: e.g., $a(bc)$ is equivalent to $a \circ (b \circ c)$. It is not part of the definition of a group, but it follows directly from the axioms that the identity and inverses are unique.

Example 1. The integers $\mathbb{Z}$ equipped with addition is a group with the identity 0.

Example 2. The integers $\mathbb{Z}$ equipped with multiplication is **not** a group – 1 would have to be the identity, but there is no integer n such that $n \cdot 0 = 1$. However, $\mathbb{Z} \setminus \{0\}$ with multiplication is a group.

Example 3. The integers $\mathbb{Z}$ modulo n, notated $\mathbb{Z}/n\mathbb{Z}$ and consisting of the integers $\{0, 1, \ldots, n-1\}$ equipped with addition modulo n are a group (and are useful for analyzing cyclic symmetry groups, as we will find later).

Often we want to demonstrate that certain groups are "the same" – that is, that they share the same underlying structure. The principal mechanism through which we can demonstrate that two groups are equivalent is via group isomorphisms.

Definition 2. *A map $\varphi : G \to H$ between two groups $(G, \circ)$ and $(H, *)$ is a **group homomorphism** if it respects the binary operation of both groups; that is, for a and b in G, it is always true that*

$$\varphi(a \circ b) = \varphi(a) * \varphi(b).$$

From this, it immediately follows that $\varphi(e_G) = e_H$, where e_G and e_H are the identity elements of G and H respectively.

*φ is a **group isomorphism** if it is also a bijection – if such a φ exists, we then say that G and H are **isomorphic** (notated $G \cong H$).*

3 Symmetry groups of the plane

One valuable application of groups can be found in studying symmetries of geometric objects. We will focus our study on the symmetry groups which manifest in finite figures on the Euclidean plane $\mathbb{R}^2$.

Definition 3. *The **symmetry group** of an object is the group whose elements are transformations (i.e., bijections) under which the object is unchanged (or invariant).*

Symmetries can be broadly classified into **reflectional**, **rotational**, **translational**, and **glide reflection** symmetries; the latter two only occur in objects which are in some way "infinite," such as tilings of the plane.

3.1 Cyclic groups

The first category of symmetry groups we will consider is those which describe objects with exclusively rotational symmetry. These symmetry groups exhibit a cyclic group structure.

Definition 4. *A group G is **cyclic** if there is an element $g \in G$ such that $G = \left\{\ldots, g^{-2}, g^{-1}, e, g, g^2, \ldots\right\}$ (where $g^2 = g \circ g$, $g^3 = g \circ g \circ g$, etc., and $g^0 = e$ by definition). g is called the **generator** of G, which can be indicated by instead notating G as $\langle g \rangle$.*

Definition 5. *The **order** $|g|$ of an element g of a group G is the smallest positive integer n such that $g^n = e$. If no such integer exists, we say that $|g| = \infty$. The order $|G|$ of the containing group is simply the number of elements of the underlying set.*

An important result is that the structure of a cyclic group is completely determined by the order of its generator and can be understood as being "the same" as one of two very simple groups.

Theorem 1. *Let $G = \langle g \rangle$ be a cyclic group with generator g.*

- *If $|g| = n$ for some integer n, then $G \cong \mathbb{Z}/n\mathbb{Z}$.*
- *If $|g| = \infty$, then $G \cong \mathbb{Z}$.*

Proof. We will consider each case in turn.

- Case 1: $|g| = n$

  First observe that $G = \left\{e, g, \ldots, g^{n-1}\right\}$ has order n, the same as $\mathbb{Z}/n\mathbb{Z}$, so they are at least compatible in terms of order.

  Then observe that since $g^n = e$, $g^{n+1} = g$, $g^{n+2} = g^2$, etc. – that is, $g^k = g^{k \pmod n}$. This yields the quite natural isomorphism $\varphi : \mathbb{Z}/n\mathbb{Z} \to G$ defined by $\varphi(k) = g^k$. To check that this respects the underlying group structure, we can see that

  $$\begin{aligned}\varphi(j + k \pmod n) &= g^{j+k \pmod n} \\ &= g^j g^k = \varphi(j)\varphi(k).\end{aligned}$$

  Thus we have an isomorphism between G and $\mathbb{Z}/n\mathbb{Z}$.

- Case 2: $|g| = \infty$

  This is even simpler. Since $g^j = g^k$ if and only if $j = k$ (by the definition of order), the map φ defined by $\varphi(k) = g^k$ is clearly an isomorphism.

□

Identifying that finite cyclic groups are "the same" as the integers modulo n offers an incredibly simple abstraction of the observation that, for instance, rotating an object by 180 degrees is the same as rotating it by 540 degrees; indeed, this property lends cyclic groups their name.

Returning to the original discussion of rotational symmetry groups, figures which are invariant under finitely many rotations are said to have n-fold symmetry:

Definition 6. *An object has n-**fold rotational symmetry** if it is invariant under every rotation by $360/n$ degrees; for instance, a figure with 3-fold rotational symmetry appears identical when rotated by 120 and 240 degrees.*

Some examples of figures with only rotational symmetry are shown in Figure 1.

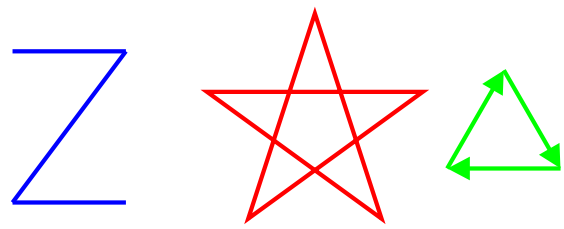

Figure 1: The letter "Z" has 2-fold rotational symmetry, a pentagram has 5-fold rotational symmetry, and the recycling symbol has 3-fold rotational symmetry.

Definition 7. *The cyclic group of all rotations by $360/n$ degrees for a positive integer n is denoted as C_n. Each C_n consists of the members*

$$\left\{R_0, R_{360/n}, R_{2\cdot 360/n}, \cdots, R_{(n-1)\cdot 360/n}\right\},$$

where each R_i denotes a rotation by i degrees. Clearly R_0 is the identity, and composition is defined quite intuitively as

$$R_{i\cdot 360/n} \circ R_{j\cdot 360/n} = R_{(i+j \pmod n)\cdot 360/n}.$$

Note that $|C_n| = n$ and that C_n is a commutative group. Objects with n-fold rotational symmetry are sometimes also simply described as having "C_n symmetry."

Example 4. $C_4 = \{R_0, R_{90}, R_{180}, R_{270}\}$.

3.2 Dihedral groups

The other category of symmetry groups we will consider is those which describe objects with both rotational and reflectional symmetry, which are modeled by the *dihedral groups*. Indeed, dihedral and cyclic groups are the *only* symmetry groups for finite figures in the plane.

Definition 8. *The **dihedral group** D_n for a positive integer n is the group of transformations consisting of*

- *n rotations $R_0, R_1, \ldots, R_{n-1}$ with R_i indicating a rotation of $i \cdot 360/n$ degrees and*
- *n reflections $S_0, \ldots, S_{n-1}$ across distinct axes of symmetry*

For $n \geq 3$, the dihedral groups are the symmetry groups of regular n-gons.

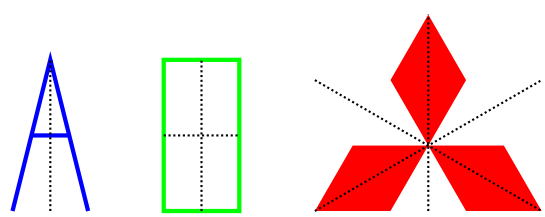

Figure 2: The letter "A" exhibits D_1 symmetry, a non-square rectangle exhibits D_2 symmetry, and the Mitsubishi logo exhibits D_3 symmetry. The dotted lines indicate the axes of symmetry for each figure.

Composition of these transformations follows a fairly straightforward and also rather suggestive pattern:

$R_i \circ R_j = R_{i+j \pmod n}$	*A rotation followed by a rotation is a rotation*
$R_i \circ S_j = S_{i+j \pmod n}$	*A reflection followed by a rotation is a reflection*
$S_i \circ R_j = S_{i-j \pmod n}$	*A rotation followed by a reflection is a reflection*
$S_i \circ S_j = R_{i-j \pmod n}$	*A reflection followed by a reflection is a rotation*

Thus D_n is not a commutative group, but its elements possess certain patterns of "evenness" or "oddness" with deeper geometric implications.

Some examples of figures which are not regular polygons but exhibit dihedral symmetry are depicted in Figure 2.

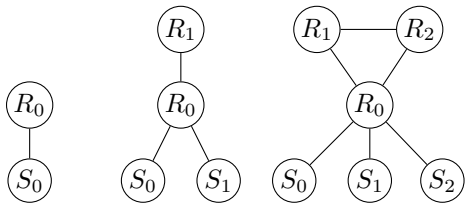

Figure 3: The cycle graphs for D_1, D_2, and D_3 demonstrate the group structure of the dihedral groups.

This group structure provides an illustration of the fact that there are no figures on the plane which have more than one axis of symmetry but no rotational symmetry – the only dihedral group with only reflectional symmetry is $D_1 = \{R_0, S_0\}$. This is a direct consequence of the fact that the composition of two reflections is equivalent to a rotation.

We can also observe that each dihedral group D_n has as a subgroup the cyclic group C_n – the combination of the non-cyclic reflections with the cyclic rotation subgroup is the defining characteristic of the dihedral groups, as illustrated for small n in Figure 3.

4 Representations and group actions

We often want to understand how these abstract groups of transformations can be used to model transformations on another, less abstract space – one powerful tool that can allow us to do this is the use of *group representations* and *group actions*. For our purposes, we will examine group representations and actions which allow symmetry groups to perform transformations of the Euclidean plane.

Definition 9. *Let X be any set. The **automorphism group** $\mathrm{Aut}(X)$ is the group consisting of all bijections from X to itself under function composition.*

Since the Euclidean plane can be understood as the real vector space $\mathbb{R}^2$, $\mathrm{Aut}(\mathbb{R}^2)$ is more specifically the space of *linear transformations* from $\mathbb{R}^2$ to itself, notated as the *general linear group* $\mathrm{GL}(\mathbb{R}^2)$. This will become significant as we study representations of symmetry groups, as this allows us to represent all symmetry groups of objects on the plane through ordinary 2×2 matrices.

Definition 10. *Let G be a group. A **representation of** G **on** $\mathbb{R}^2$ is a group homomorphism $G \to \mathrm{GL}(\mathbb{R}^2)$. That is, $\varphi : G \to \mathrm{GL}(\mathbb{R}^2)$ is a representation of G on $\mathbb{R}^2$ if*

- $\varphi(e) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\varphi(g \circ h) = \varphi(g)\varphi(h)$ *for* $g, h \in G$

Group representations are closely related to group *actions*, which can be understood as the application of a group representation to the underlying target set.

Definition 11. *Let G be a group and X be any set. The function $\varphi : G \times X \to X$ is a **group action** if*

- $\varphi(e, x) = x$ *for all* $x \in X$ *and*
- $\varphi(g, \varphi(h, x)) = \varphi(gh, x)$ *for all* $g, h \in G$ *and* $x \in X$.

*We say that G **acts on** X.*

The following proposition is trivial enough to not require a proof, but it illustrates the close relation between group actions and group representations.

Proposition 1. *Let $\varphi : G \times \mathbb{R}^2 \to \mathbb{R}^2$ be a group action. Then there is a unique group representation $\psi : G \to \mathrm{GL}(\mathbb{R}^2)$ such that $\varphi(g, x) = \psi(g)(x)$ for all $g \in G$ and all $x \in \mathbb{R}^2$.*

4.1 Example

We will construct a representation φ of D_4 on $\mathbb{R}^2$ based on the symmetries of a square with vertices at $(-1,1),(1,1),(1,-1),(-1,-1)$. This essentially requires finding the 8 2×2 matrices under which this square is invariant and assigning each to a member of D_4 in a way that respects the underlying group structure. Unsurprisingly, this means that reflections are mapped to matrices which represent reflections of the plane and rotations are mapped to matrices which represent rotations of the plane, as follows. The effect of this representation is depicted in Figure 4.

$$\varphi(R_0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \varphi(S_0) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$
$$\varphi(R_1) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \qquad \varphi(S_1) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$\varphi(R_2) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \qquad \varphi(S_2) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\varphi(R_3) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \varphi(S_3) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

This can perhaps be viewed as a "canonical" representation of D_4, as it significantly also yields *isometries* of the plane and rotation matrices which correspond to rotation of the plane; however, there are infinitely many other representations of D_4 which amount effectively to composing some other invertible linear transformation with these "canonical" matrices.

Indeed, this fact indicates that *symmetry is preserved under any invertible linear transformation*, as performing an invertible linear transformation on an object is equivalent to constructing a representation of that object's symmetry group which composes that transformation with the matrices associated with its "canonical" representation (i.e., that representation which maps the members of the symmetry group to isometries and "natural" rotations).

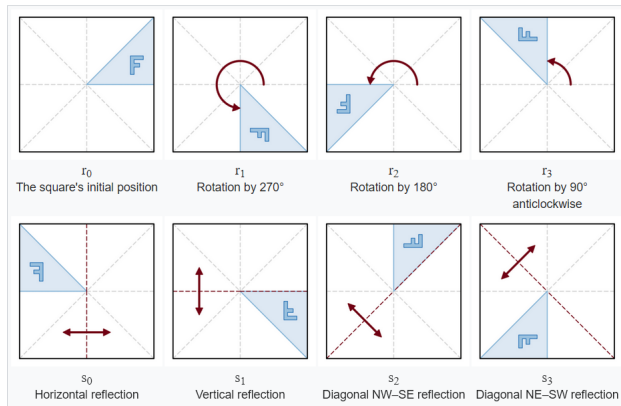

Figure 4: The effect of this representation on the letter "F." Image credit Jacob Rus on Wikimedia Commons, licensed under CC BY-SA 4.0

5 Conclusion

Symmetry groups of the plane exhibit a beautiful intersection of group theory, geometry, and linear algebra – their group theoretical aspects allow us to categorize and classify the symmetries which are possible on the plane, while the fact that their representations correspond to linear transformations allows us to justify the idea that these symmetries are preserved under familiar geometric and linear algebraic transformations.